

1) What is the primary purpose of a firewall in a network security infrastructure?

b) Filtering and controlling network traffic

2) What type of attack involves flooding a network with excessive traffic to disrupt normal operation?

a) Denial of Service (DoS)

3) Which encryption protocol is commonly used to secure wireless network communications?

b) WPA (Wi-Fi Protected Access)

4) What is the purpose of a VPN (Virtual Private Network) in a network security context?

Answer: The purpose of a VPN is to create a secure, encrypted connection over a less secure network, such as the internet, ensuring privacy and security of data transmission.

5) True or False: Patch management is the process of regularly updating software and firmware to address security vulnerabilities and improve system performance.

--> True

6) True or False: A network administrator should perform regular backups of critical data to prevent data loss in the event of hardware failures, disasters, or security breaches.

--> True

7) True or False: Traceroute is a network diagnostic tool used to identify the route and measure the latency of data packets between a source and destination device.

--> True

8) Describe the steps involved in conducting a network vulnerability Assessment?

--> The following steps are as follows

step 1) Define the scope

step 2) Gather Information

step 3) perform Vulnerability scanning

step 4) Analyze & prioritize Results

step 5) Remediation Planning

step 6) Re-scan and Verify

step 7) Report and Document

9) Demonstrate how to troubleshoot network connectivity issues using the ping command.

--> Step 1: Check Local Network Stack

Step 2: Check Your Own IP Address

Step 3: Ping the Default Gateway

Step 4: Ping an External IP

Step 5: Ping a Domain Name

Note: Interpreting Results

Reply received: Network is reachable.

Request timed out / destination unreachable: Something is blocking communication (firewall, routing, cable, etc.).

High latency (time > 100ms): Network congestion or poor connection quality.

10) Discuss the importance of regular network maintenance and the key tasks involved in maintaining network infrastructure.

-->Importance of Regular Network Maintenance

1. Improved Security

Keeps devices updated with the latest patches.

Prevents exploitation of known vulnerabilities.

Helps in detecting and removing unauthorized devices or connections.

2. Enhanced Performance

Identifies and resolves bottlenecks.

Optimizes bandwidth usage.

Reduces downtime and slow response times.

3. Increased Reliability

Prevents system failures and interruptions.

Ensures consistent connectivity.

Minimizes the risk of data loss and hardware breakdowns.

4. Scalability and Planning

Helps monitor usage trends.

Prepares the infrastructure for future upgrades or expansion.

→ Key Tasks in Network Maintenance

1. Regular Software and Firmware Updates

Apply patches to routers, switches, firewalls, and endpoints.

Ensure antivirus and security software are up to date.

2. Monitoring and Performance Checks

Use network monitoring tools (e.g., Nagios, SolarWinds, PRTG).

Track bandwidth, uptime, and response times.

Check for unusual traffic patterns or errors.

3. Backup and Recovery

Schedule regular backups of configurations and data.\

Test restoration procedures to ensure disaster recovery readiness.

4. Hardware Inspection

Check physical components (cables, ports, devices) for wear and tear.

Replace faulty equipment before failure.

5. Security Audits

Scan for vulnerabilities and unauthorized access.

Review firewall and access control policies.

Conduct penetration testing if needed.

6. Documentation and Configuration Management

Maintain accurate records of IP addresses, device configurations, and network topology.

Update network diagrams and documentation after changes.

7. User Access and Account Reviews

Disable unused or outdated accounts.

Enforce strong password and authentication policies.

Review permissions and access rights periodically.